

Decouple and Synchronize: Multi-Agent Cooperation for Adaptive Curriculum Optimization

Anonymous Authors¹

Abstract

Automatic Curriculum Learning (ACL) explores the environment parameter space to generate adaptive and smooth curricula, which is promising for scaling Reinforcement Learning (RL) to complex tasks. However, the curriculum quality is validated via policy performance, which in turn depends on the curriculum difficulty, leading to an inherent entanglement between the evaluation of curriculum and policy. Besides, ACL progress is uncontrollable and easily diverges from the target task. To address these challenges, we propose a novel ACL framework via Multi-Agent policy Decoupling and Synchronization (MADS). MADS models ACL and policy optimization as two-agent cooperation and applies multi-agent credit assignment to decouple the impact of the curriculum agent. It further facilitates curriculum convergence toward the target via dual policy synchronization, which aligns the curriculum-guided policy with a target policy (anchored on the target curriculum) through periodic parameter sharing and feedback adjustment. Empirical results demonstrate that MADS effectively accelerates policy learning. MADS overcomes the challenges of curriculum evaluation and degeneration, offering a novel perspective on ACO.

1. Introduction

Reinforcement Learning (RL) has demonstrated remarkable potential in sequential decision-making problems, yet its scalability to complex tasks is severely constrained by the inefficiency of the trial-and-error paradigm. Curriculum Learning (CL) emerges as a key technique to address this limitation by designing task sequences from easy to hard to guide learning. Particularly, Automatic Curriculum Learn-

¹Anonymous Institution, Anonymous City, Anonymous Region, Anonymous Country. Correspondence to: Anonymous Author <anon.email@domain.com>.

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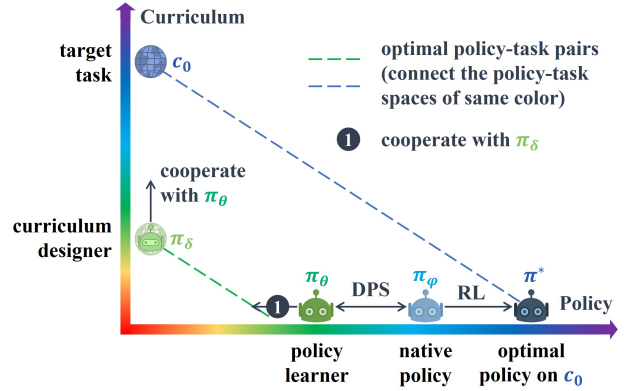


Figure 1. Optimization of policy learner π_θ and curriculum designer π_δ in the hypothetical policy-curriculum space. π_θ and π_δ cooperate to align with the dashed lines (optimal cooperation). The native policy π_ϕ gradually converges to the optimal policy π^* on the target task c_0 through RL. We introduce Dual Policy Synchronization (DPS) to drive the dual policies π_θ and π_ϕ toward each other. The native policy provides a task anchor for the policy learner, while the policy learner shares the skills learned in the curriculum with the native policy.

ing (ACL) moves beyond rigid, hand-crafted curricula by autonomously exploring the environment parameter space to generate adaptive, smooth curriculum sequences. By structuring learning into manageable and progressive stages, ACL promises to accelerate convergence and enhance generalization in challenging RL scenarios.

Current ACL methods primarily focus on defining curriculum difficulty criteria and designing progress schedules. The former introduces domain-specific metrics (e.g., data length and complexity), model-driven dynamic indicators (e.g., model loss), and supplementary factors (e.g., sample diversity). These criteria mainly rely on external signals, yet overlook that policy performance is the most essential validation of curriculum quality. The latter proposes difficulty-increasing scheduling strategies, curriculum sampling (e.g., according to the model’s current competence) and guiding (e.g., teacher-student frameworks) mechanisms. Such strategies and mechanisms either rely on hand-crafted rules to constrain curriculum direction, or suffers from curriculum degeneration. Curriculum degeneration refers to the

over-simplification of curricula to pursue short-term gains thus deviating from the ultimate goal.

To achieve effective curriculum evaluation and adaptive progress scheduling, we propose a novel ACL framework via Multi-Agent policy Decoupling and Synchronization (MADS). MADS models curriculum design and policy learning as a two-agent fully cooperative problem. We introduce a multi-agent credit assignment approach to decouple the evaluation of the curriculum designer and policy learner from the shared environmental reward, which eliminates additional task-specific criteria. Additionally, since the environmental reward serves as the most essential evaluation metric for policy performance, such decoupling avoids estimation bias compared to the hand-crafted criteria. Building on this cooperative setup, we further propose dual policy synchronization to guide curriculum alignment. Our core insight is: by introducing an extra dual policy that shares parameters with the policy learner and is anchored on the target task, the curriculum autonomously converges to the target curriculum while optimizing two-agent cooperation. Fig. 1 visualizes this insight in a hypothetical policy-curriculum space.

Specifically, MADS adopts a Centralized Training with Decentralized Execution (CTDE) paradigm. The decentralized execution models curriculum designer and policy learner as two agents with independent decision-making processes, while the centralized training leverages shared environmental rewards and global states to accommodate the temporal scale difference between the two agents. Together, the CTDE paradigm enables evaluation decoupling via monotonic value decomposition. For training stability, curriculum designer can not change the environment frequently. The value decomposition structure inherently accommodates the temporal scale difference between the two agents, decoupling sufficient training for the policy learner on dense state samples and curriculum feature extraction for the curriculum designer from sparse curriculum samples. Since the curriculum designer would simplify the task to cater to the policy learner, the two-agent cooperative paradigm is susceptible to curriculum degeneration. To address this problem, we introduce dual policy synchronization, which regularly synchronizes parameters between the learner’s policy and an additional policy trained under the target curriculum.

The key contributions of this paper are: (1) we introduce multi-agent credit assignment to decouple the temporal different curriculum and policy evaluation; (2) We propose dual policy synchronization to address the curriculum degeneration under the two-agent cooperation framework. (3) MADS is empirically demonstrated to accelerate the RL training automatically and effectively.

2. Backgrounds

2.1. Reinforcement Learning

Reinforcement Learning (RL) provides a framework for sequential decision-making in which an agent learns through interaction with an environment (?). RL is formalized as a Markov Decision Process (MDP) $(\mathcal{S}, \mathcal{A}, p, r, \gamma)$, where $\mathcal{S}$ and $\mathcal{A}$ denote the state and action spaces. The transition kernel $p(s' | s, a)$ defines the environment dynamics, and the reward function $r(s, a)$ assesses the desirability of taking action a in state s . The agent seeks a policy $\pi(a | s)$ that maximizes the discounted return,

$$J(\pi) = \mathbb{E}_{\pi, p} \left[\sum_{t=0}^{\infty} \gamma^t r(s_t, a_t) \right].$$

Modern policy optimization, including actor-critic methods, relies on estimating value functions (such as $V^\pi(s)$ or $Q^\pi(s, a)$) to guide policy updates. Although these methods achieve strong performance in many domains, RL often requires extensive interaction data, and learning can be slow or unstable when the training distribution of tasks is poorly aligned with the agent’s current ability level. This motivates the use of curriculum learning strategies that adapt task selection to facilitate efficient policy improvement.

2.2. Automatic Curriculum Learning

Many RL environments provide a parameterized family of tasks, and selecting suitable tasks during training can greatly improve sample efficiency. Automatic Curriculum Learning (ACL) studies how to choose tasks that match the learner’s evolving abilities.

Existing ACL approaches differ mainly in how they define curriculum signals. One line estimates learning progress or competence improvement, including RIAC, SAGG-RIAC, and continuous-parameter methods such as ALP-GMM (?) and GoalGAN (?). These methods adapt task sampling based on progress statistics, yet progress is summarized by a single scalar that does not distinguish policy capability from intrinsic task difficulty.

A second line uses value-function information, relying on quantities such as temporal-difference error, value uncertainty, or ensemble disagreement. Value Disagreement (?) selects goals with high uncertainty among Q-functions, and Prioritized Level Replay (??) samples environment instances with large temporal-difference errors. These approaches identify challenging tasks more precisely but still reduce curriculum utility to a single scalar value.

A third line generates full environments to induce learning. Unsupervised Environment Design methods such as PAIRED (?) produce challenging but solvable environments by optimizing regret between learners, with later

work improving stability through regularization, replay, or co-learnability metrics (???). These approaches provide rich curricula but primarily rely on adversarial objectives and do not separate the influence of the curriculum from that of the policy.

Across all families, curriculum quality is ultimately evaluated through a single scalar signal such as progress, temporal-difference error, or regret. This scalarization mixes task difficulty with policy capability, obscures teacher-student credit assignment, and often yields curriculum degeneration where short-term improvement dominates long-term alignment with the target task. In this work, we address these limitations by framing curriculum generation as a cooperative multi-agent problem with explicit value decomposition and structured policy representation, enabling curriculum selection that is stable, interpretable, and aligned with the target distribution.

3. Methodology

In this section, we introduce the MADS framework, which consists of two core components illustrated in Fig.2.

3.1. Multi-Agent Credit Assignment for Utility Decoupling

Problem Formulation. The goal of curriculum optimization is to guide the training of the policy learner, enabling it to gradually adapt to the target task. As the policy and curriculum are optimized in parallel, this process can be framed as a fully cooperative two-agent problem. The policy learner optimizes the policy under a parameterized environment while the curriculum designer optimizes the environmental parameters to match the policy’s capability. Their common goal is to maximize the environmental reward.

Accordingly, we model the policy learner and curriculum designer as two cooperative agents operating on different temporal scales. The environment is parameterized and initiated by curriculum parameters c_0 , which is also the parameter of target task. Before each episodic rollout, the curriculum designer extract the feature of learner’s policy π_θ and takes an episodic macro action $c_{ep} \sim \pi_\delta(\cdot|\pi_\theta)$. c_{ep} is applied to replace the environmental parameters of last episode. Then the policy learner takes micro actions $a_t \sim \pi_\theta(\cdot|s_t)$ at every time step $t \in [1, T]$ under the environment defined by c_{ep} . The environment reward $r(s_t, a_t, c_{ep})$ is the sole and shared external evaluation signal of the team, which reflects the policy performance under current curriculum c_{ep} . The total expected discounted return of the team is

defined as

$$Q_{\text{tot}}^{\pi_\theta}(s_t, a_t, c_{ep}) = \mathbb{E}_{s_k, a_k \sim \pi_\theta} \left[\sum_{k=t}^T \gamma^{k-t} r(s_k, a_k, c_{ep}) \right] \quad (1)$$

We recast $Q_{\text{tot}}^{\pi_\theta}(s_t, a_t, c_{ep})$ to $Q_{\text{tot}}(s_t, a_t, c_{ep}, \pi_\theta)$, which are mathematically equivalent.

Value Decomposition. In order to decouple the contribution of action a_t and curriculum design c_{ep} , we need to decompose $Q_{\text{tot}}(s_t, a_t, c_{ep}, \pi_\theta)$ into separate value functions $Q_{\pi_\theta}(s_t, a_t)$ and $Q_{\pi_\delta}(\pi_\theta, c_{ep})$. The key to decomposing Q_{tot} is selecting an appropriate decomposition operator. Monotonic value decomposition is the most popular value decomposition method, which has demonstrated strong empirical performance in complex fully cooperative multi-agent RL tasks, as it achieves a delicate balance between representational capacity and decoupling efficiency. Leveraging monotonic operator, Q_{tot} is decomposed as

$$Q_{\text{tot}}(s_t, a_t, c_{ep}, \pi_\theta) = \omega_1(s_t) \cdot Q_{\pi_\theta}(s_t, a_t) + \omega_2(s_t) \cdot Q_{\pi_\delta}(\pi_\theta, c_{ep}) \quad (2)$$

where $\forall s_t \in S$, $\omega_1(s_t) > 0$ and $\omega_2(s_t) > 0$, ensuring the increases in Q_{π_θ} and Q_{π_δ} always improve Q_{tot} . $Q_{\pi_\theta}(s_t, a_t)$ quantifies the intrinsic value of π_θ , which is invariant to curriculum variations, while $Q_{\pi_\delta}(\pi_\theta, c_{ep})$ measures the value of curriculum c_{ep} in guiding π_θ ’s improvement.

Such decomposition follows the Centralized Training with Decentralized Execution (CTDE) paradigm, where the action selection of π_θ and curriculum design of π_δ are executed independently. Besides, such decomposition effectively addresses the temporal scale inconsistency: it allows $\omega_1(s_t) \cdot Q_{\pi_\theta}(s_t, a_t)$ and $\omega_2(s_t)$ to be trained sufficiently on dense state samples while $Q_{\pi_\delta}(\pi_\theta, c_{ep})$ focuses on extracting curriculum features from sparse curriculum samples. Note the monotonic decomposition of the total Q value function is an approximation, which simplifies and decouples the impact of curriculum designer and policy learner, providing a decentralized foundation to optimize both agents independently.

Policy Feature Extraction. Note that both π_δ and Q_{π_δ} depend on the learner’s policy π_θ , so we need to extract a compact feature representation of π_θ , denoted by $f(\pi_\theta)$. To obtain $f(\pi_\theta)$ efficiently, we expect to extract policy features without additional environment interaction. A natural attempt is to leverage historical rollouts, e.g., τ_{i-1} , where i denotes current iteration. τ_{i-1} encodes the feature of $\pi_{\theta_{i-1}}$, which yields $Q_{\pi_\delta}(f(\pi_{\theta_{i-1}}), c_{ep,i})$, measuring the adaptivity of current curriculum $c_{ep,i} \sim \pi_{\delta_i}$ to previous policy $\pi_{\theta_{i-1}}$. However, the observed reward $r(s_{t,i}, a_{t,i}, c_{ep,i})$ and consequent Q_{tot} quantify the mutual adaptivity between $c_{ep,i}$ and π_{θ_i} . As a result, historical rollouts lead to a mismatch between the actual Q_{tot} and its decomposed component Q_{π_δ} .

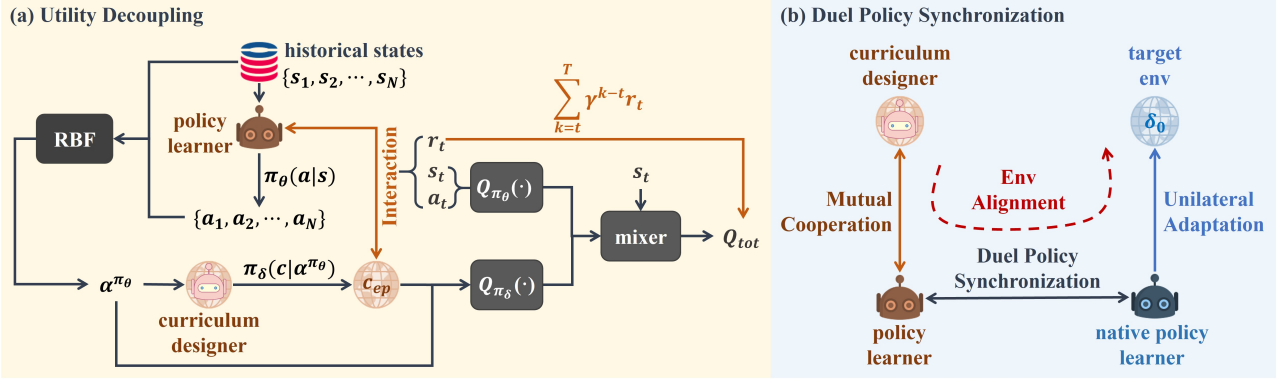


Figure 2. Core components of MADS framework. (a) Utility Decoupling decomposes the total value function Q_{tot} (derived from environmental rewards) into the time-step level, policy-specific $Q_{\pi_\theta}(s_t, a_t)$ and the episodic level, curriculum-specific $Q_{\pi_\delta}(\alpha^{\pi_\theta}, c_{ep})$. We introduce Radial Basis Function (RBF) to extract the feature of π_θ on a set of historical states, which serves as the input of curriculum designer. (b) Dual Policy Synchronization shares parameters between the policy learner and a native policy learner anchored on the target environment, enabling implicit environment alignment through indirect adaptation from curriculum designer to the native policy.

To overcome this challenge, we leverage current learner’s policy π_{θ_i} to generate greedy action $\hat{a} = \arg \max_{a \in A} \pi_{\theta_i}(a|s)$ on a set of historical states $\{s_1, s_2, \dots, s_N\}$, yielding a set of state-action pairs $\mathcal{D}_{s,a}^{\pi_{\theta_i}} = \{(s_1, \hat{a}_1), (s_2, \hat{a}_2), \dots, (s_N, \hat{a}_N)\}$. $\mathcal{D}_{s,a}^{\pi_{\theta_i}}$ encodes the feature of π_{θ_i} , which can be further compressed into $f(\pi_{\theta_i})$. For simplicity, we omit the subscript i in the following context. First, we project state s_n and action $\hat{a}_n$ ($n \in [1, N]$) to low-dimensional subspaces via predefined random matrices $\omega_s \in \mathbb{R}^{d_{proj} \times d_s}$ and $\omega_a \in \mathbb{R}^{d_{proj} \times d_a}$ respectively, where

$$\begin{aligned} s_n^{proj} &= \omega_s \cdot s_n \\ \hat{a}_n^{proj} &= \omega_a \cdot \hat{a}_n \end{aligned} \quad (3)$$

Elements of both ω_s and ω_a are sampled from $\mathcal{N}(0, \frac{1}{d_{proj}})$ to preserve the expectation of pairwise distances.

Second, we introduce Radial Basis Function (RBF) interpolation to compress the decision logic of π_{θ_i} . Unlike action frequency statistics or global value averaging that only captures statistical tendencies of policies, RBF retains accurate policy logic, ensuring qualified input for curriculum designer. Unlike inefficient distillation approaches, it also avoids extra training and compresses redundant policy parameters into a low-dimensional vector. Specifically, we adopt a Gaussian kernel to model the similarity between state s_n and s_m .

$$K(s_n^{proj}, s_m^{proj}) = \exp \left(-\frac{\|s_n^{proj} - s_m^{proj}\|^2}{2\sigma^2} \right) \quad (4)$$

where $\sigma = \frac{1}{N^2} \sum_{n=1}^N \sum_{m=1}^N \|s_n^{proj} - s_m^{proj}\|^2$ is set as the average pairwise distance of $\{s_1^{proj}, \dots, s_N^{proj}\}$, adapting the state space scale. Third, we approximate the action mapping

of π_θ as a linear combination of RBFs:

$$\hat{a}_n^{proj} \approx \sum_{m=1}^N \alpha_m \cdot K(s_n^{proj}, s_m^{proj}) \quad (5)$$

By substituting $\{(s_1^{proj}, \hat{a}_1^{proj}), (s_2^{proj}, \hat{a}_2^{proj}), \dots, (s_N^{proj}, \hat{a}_N^{proj})\}$ into Eq.5, we can solve the coefficient vector $\alpha^{\pi_\theta} = [\alpha_1, \dots, \alpha_N]^\top$ via the linear system:

$$[K(s_n^{proj}, s_m^{proj})]_{n,m=1}^N \cdot \alpha^{\pi_\theta} = [\hat{a}_1^{proj}, \hat{a}_2^{proj}, \dots, \hat{a}_N^{proj}]^\top \quad (6)$$

the kernel matrix $[K(s_n^{proj}, s_m^{proj})]_{n,m=1}^N$ is symmetric positive-definite due to the Gaussian kernel’s Mercer property, ensuring a unique and stable solution for α^{π_θ} . The coefficient vector α^{π_θ} can be directly adopted as the policy feature $f(\pi_\theta) = \alpha^{\pi_\theta}$. α^{π_θ} compactly encodes how π_θ maps states to actions, where each α_n quantifies the contribution of the n -th historical state to the overall state-action mapping.

Decoupling Loss. We further recast $\omega_2(s_t) \cdot Q_{\pi_\delta}(\alpha^{\pi_\theta}, c_{ep})$ as $\vec{\omega}_2(s_t) \cdot \vec{F}_\delta(\alpha^{\pi_\theta}, c_{ep})$, where $\vec{F}_\delta(\alpha^{\pi_\theta}, c_{ep})$ encodes the key attributes of curriculum c_{ep} and their relevance to π_θ ’s capability. $\vec{\omega}_2(s_t)$ quantifies the state value sensitivity to these features. Such recasting efficiently improves the evaluation of new curricula.

The two-agent cooperative loss is defined on the Temporal Difference (TD) error on the total Q value function.

$$\begin{aligned} \mathcal{L}_{\theta,\delta} &= \mathbb{E}_{\substack{a_t \sim \pi_\theta(\cdot|s_t) \\ c_{ep} \sim \pi_\delta(\cdot|\alpha^{\pi_\theta})}} \|Q_{tot}(s_t, a_t, c_{ep}, \pi_\theta) - r(s_t, a_t, c_{ep}) \\ &\quad - \gamma Q_{tot}(s_{t+1}, a_{t+1}, c_{ep}, \pi_\theta)\|^2 \end{aligned}$$

where $Q_{tot}(s_t, a_t, c, \pi_\theta)$

$$= \omega_1(s_t) \cdot Q_{\pi_\theta}(s_t, a_t) + \vec{\omega}_2(s_t) \cdot \vec{F}_\delta(\alpha^{\pi_\theta}, c_{ep}) \quad (7)$$

This TD error is backpropagated to update Q_{π_θ} , $\vec{F}_\delta$, and the weight functions ω_1 and $\vec{\omega}_2$.

3.2. Dual Policy Synchronization for Environment Alignment

Curriculum Degeneration. The introduced two-agent fully cooperative framework decouples the value contributions of curriculum designer and policy learner, yet is inherently susceptible to a fundamental challenge, which we term Curriculum Degeneration. In the fully cooperative multi-agent system, all agents share a team reward and cooperate with each other to maximize the expectation of accumulative rewards. From the view of curriculum designer, the goal is to cater to the policy learner’s current capability. It tends to bias the curriculum generation toward the learner’s solution, allowing the policy learner to achieve highest return with minimal effort. As a result, during the early training stages, the curriculum designer reduces the curriculum complexity. Over time, the two-agent system converges to a low-effort cooperation equilibrium, where the policy learner iterates to a simple, task-specific policy and the designer stabilizes at providing overly easy tasks. This directly deviates from the original intent of curriculum design, i.e., guiding the policy to gradually adapt to complex target tasks.

Dual Policy Synchronization. To address the curriculum degeneration, we need to balance a trade-off between two conflicting objectives: accelerating the policy learner’s progress and ensuring the curriculum aligns with the complex target task. Imprecise handling of this balance leads to opposite dilemmas. The curriculum may prioritize rapid alignment with the target task, resulting in policy learning difficulty, or it may over-cater to the policy learner, slowing the alignment process. To adaptively handle this trade-off, we introduce Dual Policy Synchronization (DPS), which achieves a self-regulating balance without explicit regularization. The core idea is to introduce a native policy learner $\pi_\phi(a|s)$ that is exclusively trained on the fixed, complex target curriculum c_0 . The original policy learner $\pi_\theta(a|s)$ remains cooperative with the curriculum designer $\pi_\delta(c_{ep}|\alpha^{\pi_\theta})$ under the dynamic curriculum c_{ep} . These two policy learners are designed to periodically and partially share parameters as

$$\begin{aligned}\theta &= \lambda_1 \theta + (1 - \lambda_1) \phi \\ \phi &= \lambda_2 \phi + (1 - \lambda_2) \theta\end{aligned}\quad (8)$$

where $\lambda_1, \lambda_2 \in (0, 1)$ determine the degree of parameter sharing. Such bi-directional parameter sharing enables a smooth curriculum alignment while adaptive curriculum design according to the capability of current policy. The mechanism relies on two complementary knowledge transfers.

Design of λ . λ_1 modulates skill transfer from π_θ to π_ϕ .

The original policy π_θ quickly acquires fundamental skills within the simpler, adaptively structured curriculum c_{ep} . By sharing parameters via the update $\theta \rightarrow \phi$, π_θ transfers these skills to the native policy π_ϕ , accelerating the learning of π_ϕ on the challenging target task. We define

$$\lambda_1 = \frac{A_{\theta \rightarrow \phi}(\tau_\theta)}{A_{\theta \rightarrow \phi}(\tau_\theta) + A_0} \quad (9)$$

where $A_0 > 0$ is a hyper-parameter, and

$$A_{\theta \rightarrow \phi}(\tau_\theta) = \max \left(0, \frac{1}{T} \sum_{s_t, a_t}^{\tau_\theta} [Q_{\pi_\theta}(s_t, a_t) - V_{\pi_\phi}(s_t)] \right) \quad (10)$$

$V_{\pi_\phi}(s_t) = \frac{1}{N} \sum_{n=1}^N Q_{\pi_\phi}(s_t, a_n)$ denotes the state value of policy π_ϕ under state s_t , where $a_n \sim \pi_\phi(a|s_t)$ is an action sample. T denotes the length of trajectory τ_θ .

λ_2 modulates alignment guidance from π_ϕ to π_θ . The native policy π_ϕ continuously accumulates target task experience from the fixed target task c_0 . Through the parameter update $\phi \rightarrow \theta$, π_ϕ transfers the target experience to the original policy π_θ , which cooperates with the curriculum designer through the two-agent cooperation framework. As a result, to incorporate the experience from π_ϕ , the curriculum designer is implicitly guided to adjust the dynamic curriculum toward c_0 . We therefore define

$$\lambda_2 = \exp \left(-D \left(\vec{F}_\delta(\alpha^{\pi_\theta}, c_{ep}), \vec{F}_\delta(\alpha^{\pi_\theta}, c_0) \right) \right) \quad (11)$$

where D denotes a distance metric. $\vec{F}_\delta(\alpha^{\pi_\theta}, c_0)$ is the decomposition component of Eq.7 corresponding to the target curriculum c_0 . λ_2 focuses on the similarity in curriculum feature patterns rather than numerical differences.

Curriculum Sample Filtering. To balance exploration and exploitation in curriculum sampling, we sample curricula based on the curriculum designer’s policy $\pi_\delta(c|\alpha^{\pi_\theta})$ via ϵ -greedy strategy. Meanwhile, to ensure the stability of training, curricula can not be updated frequently, resulting in the sparsity of curriculum samples. Due to such sparsity, an accidentally sampled poor curriculum can continuously disrupt training over multiple iterations. Therefore, we introduce curriculum sample filtering to mitigate the uncertainty.

First, we generate a candidate curriculum set $\{c_1, c_2, \dots, c_n\}$ by sampling from $\pi_\delta(c|\alpha^{\pi_\theta})$ via ϵ -greedy strategy. The filtering process evaluates each c_n ($m \in [1, M]$) from the aspects of target alignment and return gain. The alignment to the target curriculum is measured by

$$\Delta D_{\vec{F}}(c_n) = D \left(\vec{F}_\delta(\alpha^{\pi_\theta}, c_n), \vec{F}_\delta(\alpha^{\pi_\theta}, c_0) \right) \quad (12)$$

where c_{old} is the previous curriculum, c_0 is the target cur-

riculum. The return gain is measured by

$$\begin{aligned} & \Delta Q_{\text{tot}}(c_n) \\ &= \sum_{t=1}^T [Q_{\text{tot}}(s_t, a_t, c_n, \pi_\theta) - Q_{\text{tot}}(s_t, a_t, c_{\text{old}}, \pi_\theta)] \quad (13) \\ &= \vec{\omega}_2(s_t) \left(\vec{F}_\delta(\alpha^{\pi_\theta}, c_{ep}) - \vec{F}_\delta(\alpha^{\pi_\theta}, c_{old}) \right) \end{aligned}$$

We select c_{new} from $\{c_1, c_2, \dots, c_N\}$ by

$$c_{new} = \begin{cases} \arg \max_{c_i} \Delta Q_{\text{tot}}(c_n), & \text{if } \Delta \bar{Q}_{tot} > 0 \\ \arg \min_{c_i} \Delta D_{\vec{F}}(c_n), & \text{otherwise} \end{cases} \quad (14)$$

where $\Delta \bar{Q}_{tot} = \frac{1}{N} \sum_{n=1}^N \Delta Q_{\text{tot}}(c_n)$. We use a scenario-aware selection rule based on the candidate set's average return gain $\Delta \bar{Q}_{tot}$, which prioritizes maximizing ΔQ_{tot} to accelerate policy progress when most candidates are beneficial ($\Delta \bar{Q}_{tot} > 0$), and focuses on minimizing $\Delta D_{\vec{F}}$ to maintain alignment with c_0 when candidates are low-quality ($\Delta \bar{Q}_{tot} \leq 0$). Such filtering not only reduces the uncertainty caused by sparse curriculum samples but also provides high quality curriculum inputs for the dual-policy collaboration of DPS at the sample level.

A. Related Work on Automatic Curriculum Learning

Automatic Curriculum Learning (ACL) seeks to improve reinforcement learning by selecting tasks, goals, or environment parameters that match the learner’s evolving capabilities. Existing ACL approaches differ primarily in how the curriculum signal is defined and how the teacher interacts with the student. Following the taxonomy of ? and subsequent benchmark studies such as ?, we summarize ACL methods into three broad families: (i) performance- and progress-based curricula, (ii) value- and uncertainty-driven curricula, and (iii) environment-generating or adversarial teachers.

Performance- and Progress-Based Curricula. These methods estimate the learner’s progress on tasks and select those that maximize expected improvement. Early intrinsically motivated exploration approaches such as RIAC and SAGG-RIAC measure competence progress in goal space and favor regions with high progress. ALP-GMM (?) extends this idea to continuous parameterized tasks by fitting a Gaussian mixture model over task parameters and estimated learning progress, allowing the teacher to sample tasks from high-progress components. GoalGAN (?) uses a generative model to propose “intermediate difficulty” goals, whereas reverse curriculum generation (?) expands from solvable initial states into increasingly difficult regions. More recent meta-ACL methods (??) learn teachers across multiple students, enabling curriculum knowledge transfer and amortizing curriculum search. These approaches are simple and widely applicable, but they rely on *scalar* performance or progress estimates that do not disentangle curriculum utility from the inherent difficulty of the tasks or from the evolving capabilities of the learner. As a result, they may suffer from curriculum degeneration, where the teacher repeatedly selects tasks that maximize immediate progress but deviate from the target distribution.

Value- and Uncertainty-Driven Curricula. A second family bases curriculum signals directly on value estimation or model uncertainty. Value Disagreement (?) trains an ensemble of Q-functions and selects goals where ensemble disagreement is high, corresponding to regions of high epistemic uncertainty. Prioritized Level Replay (PLR) (??) applies a similar idea to procedurally generated environments by maintaining a replay set of levels and sampling those with large temporal-difference (TD) errors, encouraging training on challenging but learnable levels. Variants of value-based ACL have been explored in zero-sum or multi-goal settings, where value variance or disagreement defines a progress signal (?). These methods can capture finer-grained difficulty information than simple reward progress, but the curriculum signal remains a single scalar per task,

implicitly blending task difficulty, policy quality, and value estimation noise. Consequently, they do not explicitly resolve the credit assignment problem between teacher and student.

Environment-Generating and Adversarial Teachers. Unsupervised Environment Design (UED) extends ACL beyond task selection to full environment generation. PAIRED (?) frames curriculum generation as a three-agent game involving a protagonist, an antagonist, and an adversarial designer; the designer is rewarded by the regret between protagonist and antagonist returns, driving it to produce challenging yet solvable environments. Subsequent work highlights instability issues in UED and introduces regularization, replay-based stabilization, or behavioral cloning to improve robustness (??). Co-learnability metrics have been proposed to quantify how tasks interact in multi-task settings and inform environment generation (?). These approaches target open-endedness and robustness across diverse tasks, but they typically adopt adversarial formulations where curriculum design is framed as a minimax problem, rather than as cooperative guidance. Moreover, UED focuses on environment-level generalization rather than on disentangling curriculum and policy contributions under a shared reward structure.

Relation to our framework. Across all three families, ACL methods rely on *scalar* curriculum signals—such as learning progress, TD error, uncertainty, or regret—to determine which tasks to select. This scalarization conflates curriculum quality, policy capability, and estimation noise, making it difficult to determine how much of the observed return should be attributed to the teacher versus the student. As a result, prior ACL methods provide limited control over curriculum degeneration and lack a principled mechanism for credit assignment. In contrast, our approach frames ACL as a cooperative multi-agent problem, introduces a structured policy representation to inform curriculum decisions, and employs value decomposition to explicitly separate teacher and student contributions under a shared reward. Dual-policy synchronization further stabilizes the learning dynamics and aligns the curriculum with the target task, addressing limitations not resolved by prior ACL families.